ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains why one factor is changed.
2	**(c)** A is true but R is false.	A is true; R is false because animal cells lack a cell wall.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** greater	12 degrees C versus 7 degrees C.
6	**(a)** 2 mm	2 x 1000 = 2000 micrometres = 2 mm.
7	**(d)** 70	40 + 30 = 70.
8	**(b)** B gets hotter than A	Thin wire has higher resistance.
9	**(b)** 50 degrees C	Subtract zero error of 3.
10	**(c)** stops functioning properly	Nucleus controls the cell.
11	**(b)** colds are viral and often recover by themselves	Colds go away without them.
12	**(c)** the iron rod	Iron concentrates the field.
13	**(c)** A is true but R is false.	A is true; R is false, repetition checks reliability but cannot guarantee truth.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Chloroplasts carry out photosynthesis, which is why plants (which make their own food) have them.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Antibodies prepare the body.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	T = 34 / 20 = 1.7 s (1); Timing many swings reduces the error of reaction time (1).	1+1
17	Actual = 5 / 500 = 0.01 mm (1); = 10 micrometres (1).	1+1
18	Unvaccinated = 2000 = 25% (1); Fewer susceptible persons means less spread (1).	1+1
19	Both feet are at the same potential (1); No current passes through the bird (1).	1+1
20	Accuracy is closeness to the true value; precision is closeness of repeated readings (1); Readings may be precise but wrong due to zero error (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Only one soil type was compared (1); The conclusion over-generalises (1); Test several soils with controls (1).	1+1+1
22	Onion cells are rectangular with a cell wall; cheek cells are rounded or irregular without a wall (1); Both show a nucleus and cytoplasm; the onion cell has a large vacuole (1); Stains such as safranin or methylene blue make the parts visible (1).	1+1+1
23	Contaminated drinking water (1); Unhygienic food (1); Preventive: boiled water, hygiene, treat water tank (1).	1+1+1
24	Increase the number of turns (1); Increase the number of cells (1); Use a thicker soft iron core (1).	1+1+1
	OR: Straight line (1); Passes through the origin (1); Ratio stays constant (1).	1+1+1
25	Digestion: food vacuole (1); Excretion: contractile vacuole (1); Movement: pseudopodia (1).	1+1+1
	OR: Goitre: iodised salt (1); Rickets: milk, sunlight (1); Anaemia: leafy vegetables (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Apparatus: equal pieces of cloth, beaker, balance (1); Independent: type of fabric (1); Constants: size, water volume, soaking time (1); Measure mass before and after, tabulate gain (1); Fabric with greatest gain absorbs most (1).	1+1+1+1+1
	OR: Onion peel stained with safranin (1); Cheek cells stained with methylene blue (1); Plant: rectangular, wall, large vacuole (1); Animal: rounded, irregular, no wall (1); Both: nucleus and cytoplasm (1).	1+1+1+1+1
27	Contaminated food or water at the feast (1); Salmonella bacteria from a carrier or dirty water (1); Hands and flies spread germs (1); Ensure clean water, cooked food, hygiene (1); Treatment with antibiotics on advice; vaccination (1).	1+1+1+1+1
	OR: Independent: number of turns; dependent: pins lifted (1); Constant: cell, nail, wire (1); Procedure: 10, 20, 30, 40 turns and count pins (1); Expected: pins rise with turns (1); Conclusion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Series connection allows it to sense all current.
2	**(a)** B and C	Only light differs between B and C.
3	**(b)** 0.05 mm	5 / 100 = 0.05 mm.
4	**(b)** 75%	15000/20000 x 100 = 75%.
5	**(b)** glows dimmer	Reversed cell opposes the others.
6	**(a)** plants without fertiliser grown in same conditions	Needed for comparison.
7	**(a)** yeast is more active at warm temperature	Warmth boosts fermentation.
8	**(b)** boiling milk	Not related to vector.
9	**(a)** the current is larger	More current.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Only testable statements can be supported or rejected.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	Breeding sites are removed.
13	**(c)** A is true but R is false.	A is true; R is false because a thick wire has less resistance.
14	**(d)** water amount is not controlled	Another variable changes.
15	**(b)** makes food by photosynthesis	Chloroplasts trap light for photosynthesis.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Amoeba has only a flexible membrane (1); Plant cell has a rigid wall (1).	1+1
17	They cannot be cured completely (1); Treatment and lifestyle changes keep them in control (1).	1+1
18	Total resistance increases (1); Less current for each (1).	1+1
19	Numbers and units are clear at a glance (1); Comparison and plotting are easier (1).	1+1
20	All organisms are built of cells (1); Cells combine into tissues, organs and bodies (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Wire resistance is high (1); Less current reaches the bulb (1); Use a short thick wire (1).	1+1+1
22	Equal size sheets, same water volume (1); Measure water absorbed (1); Repeat and average (1).	1+1+1
23	Antibiotics act on bacteria (1); Fungi have different cell walls (1); Fungicides are needed (1).	1+1+1
24	Clean water and mess hygiene (1); Regular medical check-ups (1); Mosquito control and cleanliness (1).	1+1+1
	OR: The switch breaks the circuit (1); No current flows (1); But switching off the plug is also advised (1).	1+1+1
25	Outlier: 6.5 s (1); Cause: late stop or a different path (1); Mean without it = 16.4 / 4 = 4.1 s (1).	1+1+1
	OR: Virus is smaller than bacterium (1); Virus is non-cellular; bacterium is a cell (1); Bacterial diseases need antibiotics; viral diseases need vaccines/antivirals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Curiosity starts the questions (1); Careful observation gives evidence (1); Persistence helps overcome failed trials (1); Repetition ensures reliability (1); Results shared for checking (1).	1+1+1+1+1

	OR: Cells in a multicellular body divide the work among themselves (1); Nerve cells are long and branched to carry messages (1); Red blood cells carry oxygen (1); Root hair cells have long projections to absorb water (1); Specialisation makes the body work efficiently (1).	1+1+1+1+1
27	Pathogen causes disease; host harbours it (1); Vector carries it (1); Symptoms felt, immunity resists (1); Vaccine prevents; antibiotic treats bacteria (1); Epidemic is a sudden spread (1).	1+1+1+1+1
	OR: Cell about 1.5 V (1); Battery: several cells, higher voltage (1); Mains about 220 V (1); Low voltage cannot drive a harmful current through the body (1); Mains can cause fatal shock (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** The pathogen can spread through unvaccinated people	Community protection weakens.
2	**(d)** strength rises with turns	Doubling turns gives about double pins.
3	**(a)** time for wax to drop	Result measured.
4	**(b)** tissues	Tissue means many cells.
5	**(c)** isolating the patients and hygiene	Break transmission.
6	**(a)** retains magnetism	Steel is used for permanent magnets.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Honest recording of all data is needed.
8	**(a)** Both A and R are true and R is the correct explanation of A.	The space allows more haemoglobin.
9	**(d)** A is false but R is true.	A is false: diabetes is non-communicable. R is true: it is not caused by a pathogen.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Oersted's finding.
11	**(c)** 1.6 s	48 / 30 = 1.6 s.
12	**(b)** virus, bacterium, red blood cell	Smaller to larger.
13	**(c)** typhoid	Faecal contamination.
14	**(a)** melt	Total 7 A exceeds 5 A.
15	**(c)** Does 200 mL of water boil faster than 400 mL on the same burner?	Testable and specific.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Prevent contamination of water (1); Reduce fly and mosquito breeding (1).	1+1
17	Current decreases (1); The electromagnet becomes weaker (1).	1+1
18	Sum of five = 62.0 (1); Fifth = 62.0 - 49.4 = 12.6 s (1).	1+1
19	Cork consists of dead cells (1); Only the walls remain after the contents are lost (1).	1+1
20	Flu is viral so antibiotics do not work (1); Rest and fluids help the immune system (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Vaccine prevents infection; does not cure (1); Medicine treats symptoms (1); Measles has no antibiotic cure as it is viral (1).	1+1+1
22	The heater (1); It carries a larger current (1); A thick wire has low resistance and does not overheat (1).	1+1+1
23	Wanting a certain result can influence readings (1); Use blind or double checks (1); Record all readings honestly (1).	1+1+1
24	No division of labour (1); Complex functions cannot be done (1); Organisms would stay simple (1).	1+1+1
	OR: Individual: repellent, long sleeves (1); Family: nets, no stagnant water (1); Community: fogging, drain cleaning (1).	1+1+1
25	Both give deflection since current flows (1); Iron foil may change the field slightly (1); Paper has no effect (1).	1+1+1
	OR: Experiment changes a variable to see effect (1); Survey collects information without changing anything (1); e.g. fertiliser test versus asking students' favourite fruit (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Height increases up to 60 mL (1); Falls at 80 mL (1); Too much water may reduce air in the soil (1); Next test: values between 40 and 80 mL (1); Repeat with several plants (1).	1+1+1+1+1
	OR: Objective = 800 / 10 = 80x (1); Actual = 4 / 800 mm (1); = 0.005 mm (1); = 5 micrometres (1); This size suggests a bacterium (1).	1+1+1+1+1
27	First dose: slow, small rise (1); Second dose: faster, higher rise (1); Memory cells remain (1); Levels fall later (1); Booster restores (1).	1+1+1+1+1
	OR: Total = 4.5 + 0.5 + 0.3 (1); = 5.3 A (1); Exceeds 5 A, so unsuitable (1); Fuse would melt in use (1); Choose a 10 A fuse for safety (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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